Birch bookshelf

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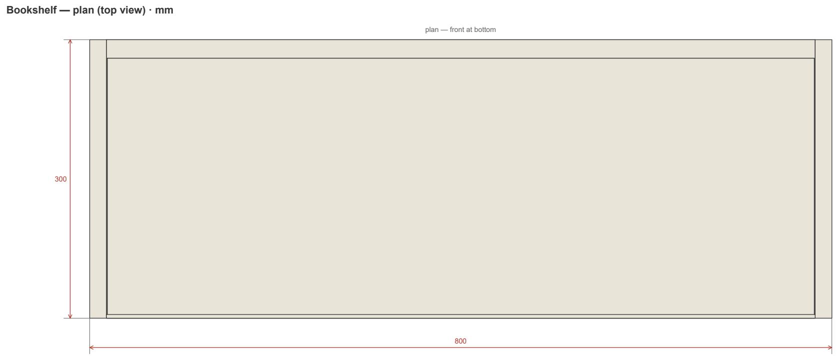
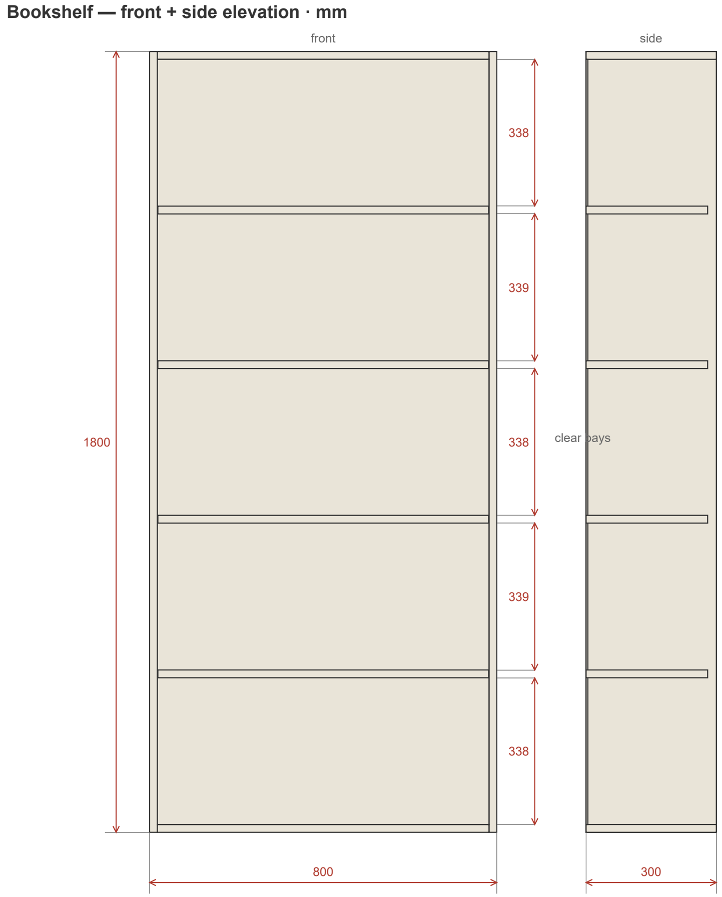
plywood_birch   hardboard

Overall: **800x1800x300** mm
Units: **mm**
Date: **2026-07-08**

Drawings



plan



front

Cut list — Birch bookshelf

Units: mm

Part	Qty	Length	Width	Thk	Material	Grain	Banding	Notes
Side	2	1800	300	18	plywood_birch 18mm / דיקט בירץ׳	length	front	
Bottom	1	764	300	18	plywood_birch 18mm / דיקט בירץ׳	length	front	
Top	1	764	300	18	plywood_birch 18mm / דיקט בירץ׳	length	front	
Back	1	1764	764	4	hardboard 4mm / מזונית	none		
Shelf	4	762	280	18	plywood_birch 18mm / דיקט בירץ׳	length	front	

Material summary

Material	Parts	Area (m^2)	Banding (m)	Sheets (est.)
plywood_birch 18mm / דיקט בירץ׳	8	2.392	8.18	1
hardboard 4mm / מזונית	1	1.348	0.0	1

Sheet count is a heuristic estimate, not an optimised cut plan. For production nesting use OpenCutList or CutList Optimizer.

Assembly Plan — Bookshelf

Overall: 800 × 1800 × 300 mm Construction style: frameless_permanent Joint method: confirmat

Tools needed

- Brad-point Ø8mm with depth stop
- Confirmat stepped drill 7×50 (Ø5mm pilot + Ø7mm clearance + countersink)
- Ø5mm brad-point
- Ø5mm with depth stop at 10mm
- Carpenter's square / angle clamp
- Clamps
- Rubber mallet
- Pencil + tape measure

Hardware list

- 4× confirmat screw 7×50mm
- 4× cover cap (optional)
- 8× fluted hardwood dowel Ø8×30mm
- 8× wood glue

Pre-assembly drilling

Do ALL drilling before you start assembling. Once panels are joined, you can't reach the inside faces.

Bottom_2

confirmat (face) — connecting to Side_0:

- **confirmat_through**: Ø7mm × through deep, at 50mm along panel, panel_centermm from edge. Bit: Confirmat stepped drill 7×50 (Ø5mm pilot + Ø7mm clearance + countersink). [confidence: high]
- **confirmat_through**: Ø7mm × through deep, at 250mm along panel, panel_centermm from edge. Bit: Confirmat stepped drill 7×50 (Ø5mm pilot + Ø7mm clearance + countersink). [confidence: high]

confirmat (face) — connecting to Side_1:

- **confirmat_through**: Ø7mm × through deep, at 50mm along panel, panel_centermm from edge. Bit: Confirmat stepped drill 7×50 (Ø5mm pilot + Ø7mm clearance + countersink). [confidence: high]
- **confirmat_through**: Ø7mm × through deep, at 250mm along panel, panel_centermm from edge. Bit: Confirmat stepped drill 7×50 (Ø5mm pilot + Ø7mm clearance + countersink). [confidence: high]

Shelf_5

glued_dowel (face) — connecting to Side_0:

- **dowel_face**: Ø8mm × 13.5mm deep, at 40mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_face**: Ø8mm × 13.5mm deep, at 240mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

glued_dowel (face) — connecting to Side_1:

- **dowel_face**: Ø8mm × 13.5mm deep, at 40mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_face**: Ø8mm × 13.5mm deep, at 240mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

Shelf_6

glued_dowel (face) — connecting to Side_0:

- **dowel_face**: Ø8mm × 13.5mm deep, at 40mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_face**: Ø8mm × 13.5mm deep, at 240mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

glued_dowel (face) — connecting to Side_1:

- **dowel_face**: Ø8mm × 13.5mm deep, at 40mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_face**: Ø8mm × 13.5mm deep, at 240mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

Shelf_7

glued_dowel (face) — connecting to Side_0:

- **dowel_face**: Ø8mm × 13.5mm deep, at 40mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_face**: Ø8mm × 13.5mm deep, at 240mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

glued_dowel (face) — connecting to Side_1:

- **dowel_face**: Ø8mm × 13.5mm deep, at 40mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_face**: Ø8mm × 13.5mm deep, at 240mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

Shelf_8

glued_dowel (face) — connecting to Side_0:

- **dowel_face**: Ø8mm × 13.5mm deep, at 40mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_face**: Ø8mm × 13.5mm deep, at 240mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

glued_dowel (face) — connecting to Side_1:

- **dowel_face**: Ø8mm × 13.5mm deep, at 40mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_face**: Ø8mm × 13.5mm deep, at 240mm along panel, panel_centermm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

Side_0

confirmat (edge) — connecting to Bottom_2:

- **confirmat_pilot**: Ø5mm × 45mm deep, at 50mm along panel, panel_thickness / 2mm from edge. Bit: Ø5mm brad-point. [confidence: high]

- **confirmat_pilot**: Ø5mm × 45mm deep, at 250mm along panel, panel_thickness / 2mm from edge. Bit: Ø5mm brad-point. [confidence: high]

confirmat (edge) — connecting to Top_3:

- **confirmat_pilot**: Ø5mm × 45mm deep, at 50mm along panel, panel_thickness / 2mm from edge. Bit: Ø5mm brad-point. [confidence: high]
- **confirmat_pilot**: Ø5mm × 45mm deep, at 250mm along panel, panel_thickness / 2mm from edge. Bit: Ø5mm brad-point. [confidence: high]

glued_dowel (edge) — connecting to Shelf_5:

- **dowel_edge**: Ø8mm × 19mm deep, at 40mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_edge**: Ø8mm × 19mm deep, at 240mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

glued_dowel (edge) — connecting to Shelf_6:

- **dowel_edge**: Ø8mm × 19mm deep, at 40mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_edge**: Ø8mm × 19mm deep, at 240mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

glued_dowel (edge) — connecting to Shelf_7:

- **dowel_edge**: Ø8mm × 19mm deep, at 40mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_edge**: Ø8mm × 19mm deep, at 240mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

glued_dowel (edge) — connecting to Shelf_8:

- **dowel_edge**: Ø8mm × 19mm deep, at 40mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_edge**: Ø8mm × 19mm deep, at 240mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

shelf_pin (face (inner)) — connecting to adjustable_shelves:

- **System-32 shelf pin rows**: 2 rows × 54 holes each
- front_row: 37mm from front edge
- rear_row: 263mm from front edge
- Hole spacing: every 32mm, starting at 37mm from bottom
- Ø5mm × 10mm deep
- Bit: Ø5mm with depth stop at 10mm
- [confidence: high]

Side_1

confirmat (edge) — connecting to Bottom_2:

- **confirmat_pilot**: Ø5mm × 45mm deep, at 50mm along panel, panel_thickness / 2mm from edge. Bit: Ø5mm brad-point. [confidence: high]
- **confirmat_pilot**: Ø5mm × 45mm deep, at 250mm along panel, panel_thickness / 2mm from edge. Bit: Ø5mm brad-point. [confidence: high]

confirmat (edge) — connecting to Top_3:

- **confirmat_pilot**: Ø5mm × 45mm deep, at 50mm along panel, panel_thickness / 2mm from edge. Bit: Ø5mm brad-point. [confidence: high]
- **confirmat_pilot**: Ø5mm × 45mm deep, at 250mm along panel, panel_thickness / 2mm from edge. Bit: Ø5mm brad-point. [confidence: high]

glued_dowel (edge) — connecting to Shelf_5:

- **dowel_edge**: Ø8mm × 19mm deep, at 40mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

- **dowel_edge**: Ø8mm × 19mm deep, at 240mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

glued_dowel (edge) — connecting to Shelf_6:

- **dowel_edge**: Ø8mm × 19mm deep, at 40mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_edge**: Ø8mm × 19mm deep, at 240mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

glued_dowel (edge) — connecting to Shelf_7:

- **dowel_edge**: Ø8mm × 19mm deep, at 40mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_edge**: Ø8mm × 19mm deep, at 240mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

glued_dowel (edge) — connecting to Shelf_8:

- **dowel_edge**: Ø8mm × 19mm deep, at 40mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]
- **dowel_edge**: Ø8mm × 19mm deep, at 240mm along panel, panel_thickness / 2mm from edge. Bit: Brad-point Ø8mm with depth stop. [confidence: high]

shelf_pin (face (inner)) — connecting to adjustable_shelves:

- **System-32 shelf pin rows**: 2 rows × 54 holes each
- front_row: 37mm from front edge
- rear_row: 263mm from front edge
- Hole spacing: every 32mm, starting at 37mm from bottom
- Ø5mm × 10mm deep
- Bit: Ø5mm with depth stop at 10mm
- [confidence: high]

Top_3

confirmat (face) — connecting to Side_0:

- **confirmat_through**: Ø7mm × through deep, at 50mm along panel, panel_centermm from edge. Bit: Confirmat stepped drill 7×50 (Ø5mm pilot + Ø7mm clearance + countersink). [confidence: high]
- **confirmat_through**: Ø7mm × through deep, at 250mm along panel, panel_centermm from edge. Bit: Confirmat stepped drill 7×50 (Ø5mm pilot + Ø7mm clearance + countersink). [confidence: high]

confirmat (face) — connecting to Side_1:

- **confirmat_through**: Ø7mm × through deep, at 50mm along panel, panel_centermm from edge. Bit: Confirmat stepped drill 7×50 (Ø5mm pilot + Ø7mm clearance + countersink). [confidence: high]
- **confirmat_through**: Ø7mm × through deep, at 250mm along panel, panel_centermm from edge. Bit: Confirmat stepped drill 7×50 (Ø5mm pilot + Ø7mm clearance + countersink). [confidence: high]

Assembly steps

Step 1: Lay down first side panel

Parts: Side_0, Side_1

1. Lay the **first** side panel flat on a clean surface, inside face up.
2. All cam housings and shelf-pin rows should already be drilled (see above).
3. The second side panel is attached **last** in this group — after bottom, dividers, and top are in place.

Step 2: Attach bottom panel

Parts: Bottom_2

1. Insert connecting bolts into the bottom panel's edge (pre-drilled holes).
2. Lower the bottom into the first side's cam housings.
3. Turn each cam 90° clockwise to lock.
4. Check square with a carpenter's square before the joint sets.

Step 3: Attach top panel

Parts: Top_3

1. Same procedure as the bottom panel.
2. After locking, measure diagonals to verify the carcass is square.

Step 4: Install back panel

Parts: Back_4

1. With the carcass lying face-down, slide the back panel into place.
2. Pin or staple at ~200mm intervals around the perimeter.
3. The back squares the carcass — measure diagonals before pinning.

Step 5: Insert adjustable shelves

Parts: Shelf_5, Shelf_6, Shelf_7, Shelf_8

1. Insert shelf pins at the desired heights in the System-32 rows.
2. Lower each shelf onto its four pins.

Final step: Wall anchoring

This unit is 1800mm tall — anchor it to the wall. any tall unit in a child's room must be anchored to the wall

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All drilling coordinates from joinery.json (verified sources). Confirm hardware-specific dimensions against the actual product you purchased.